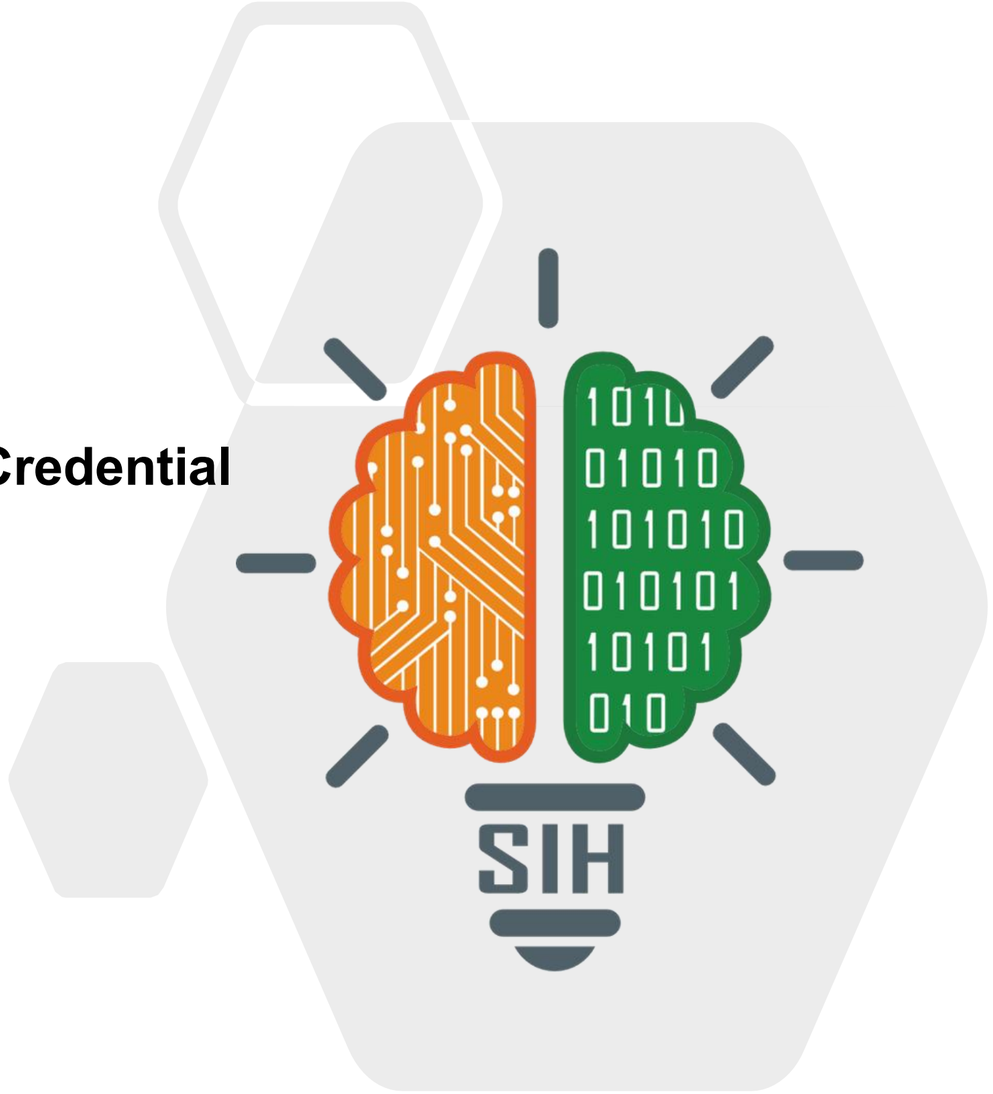
SMART INDIA HACKATHON 2026

Satyameva

- **Problem Statement ID –**
- **Problem Statement Title- Tamper Proof Academic Credential Verification System**
- **Theme- Blockchain and Cybersecurity**
- **PS Category- Software**
- **Team ID- 10**
- **Team Name (Registered on portal) - CypherX**



CypherX

Satyameva

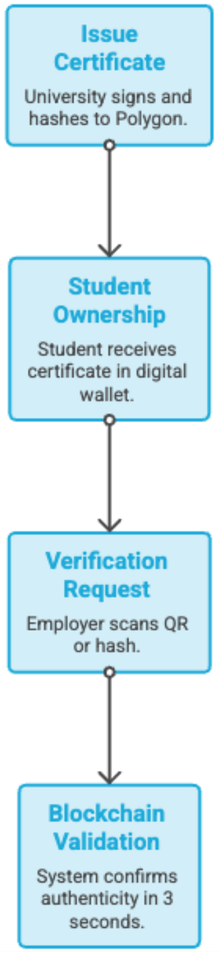
SMART INDIA HACKATHON 2026

Blockchain based certificate issuance and instant verification replacing weeks of manual checks and fake degree fraud with a 3 second, tamper-proof QR scan.

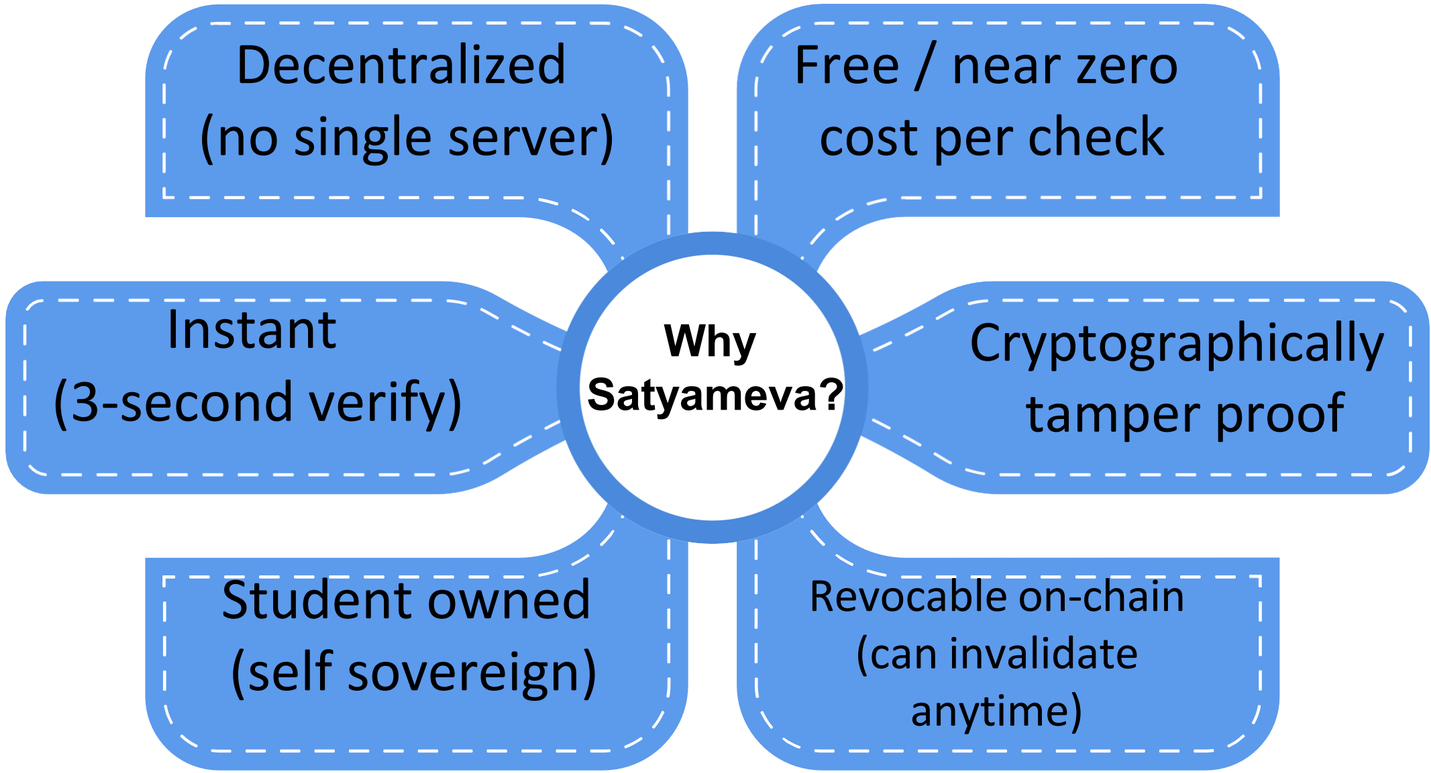
What is Satyameva?

Satyameva transforms certificates into tamper proof blockchain credentials that students own and anyone can verify instantly.

Blockchain Certificate Lifecycle



Whats the USP?



CypherX

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

Techstack and Ui mockups

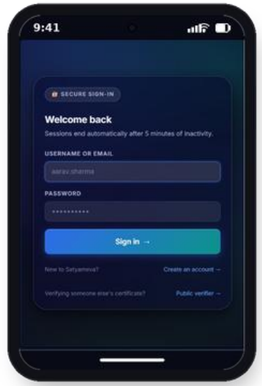
Frontend
React.js/ HTML+js

Backend
Node.js + Web3.js

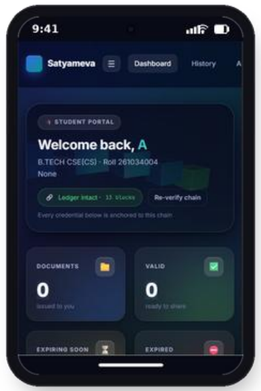
Database
SQLite, SQLAlche -my ORM

Security
SHA-256, HMAC-SHA256, PBKDF2

Blockchain
Custom PoW engine



Secure Sign-In



Student Dashboard

Methodology

Synchronous Processing → Cryptographic Tamper Proofing → Seamless Notifications

Process/ Implementation flow

1. University fills certificate details & signs digitally (private key)
2. Smart contract generates SHA-256 hash & mints it on Polygon
3. Certificate PDF is AES-256 encrypted & stored on IPFS
4. Student receives certificate + QR code in digital wallet
5. Employer scans QR / instant on-chain verification (~3 sec)

CypherX

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

~2sec
blockchain transaction / certificate mint time

₹0.08
Cost per certificate on Polygon network

₹8Lakh
total cost to issue 1 crore (10M) certificates

Phase Rollout Roadmap

Phase 1
Pilot at own college

→

Phase 2
state university network

→

Phase 3
UGC mandate+ SMS lookup

→

Phase 4
National scale
(40M+ students)

Risk and mitigation strategy

- Key compromise → Multi-sig wallet (2-of-3) + on-chain revocation
- No smartphone access - Printable QR + Phase 3 SMS lookup
- Adoption resistance → Pilot in-house, then push for UGC mandate
- DigiLocker overlap → Complementary trust layer, not a replacement

CypherX

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

40,000+
fake degrees officially caught in India every year (reported cases only)

~99.99%
faster verification: 2–4 weeks → 3 seconds

~99.99%
cheaper per check: ₹500–₹2,000 → ₹0.08

40M+
higher-education students in India (addressable scale)

Benefits Created

1. Social:Restores trust in degrees; protects genuine graduates from fraud stigma.
2. Economic: Cuts verification fees saves ~₹500-2000 Cr at 1 crore certificates.
3. Environmental: Fully paperless no mailers or couriers between institutions..
4. Policy: Aligned with NEP 2020 and Digital India.

RESEARCH AND REFERENCES